Motion Control of a Guide Dog Robot Adapted to the Intentions of Visually Impaired Individuals and Dynamic Environmental Conditions

2025 9th International Conference on Robotics and Automation Sciences (ICRAS) | 979-8-3503-9249-4/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICRAS65818.2025.11108859

Yuhui Gao
dept. of Mechanical Engineering
Kanagawa University
Yokohama, Japan
r202570142nq@jindai.jp

Bin Zhang
dept. of Mechanical Engineering
Kanagawa University
Yokohama, Japan
zhangbin@kanagawa-u.ac.jp

Abstract—To enhance the mobility and quality of life of visually impaired individuals, this paper proposed a guide dog robot by improving navigation and interaction capabilities under dynamic environments. Integrating with existing navigation technologies, two user-operated buttons are equipped to the robot to accelerate or decelerate the speed, temporarily stop or restart the robot, according to the will and operations of the user. The moving speed of the robot is finally calculated by combing the will of the user and the situation of the environment to ensure the safety of the user. Additionally, a voice interaction system is proposed to generate real-time environmental descriptions based on LSTM neural networks, conveyed through speech synthesis. Experiments demonstrate that the robot effectively responds to user commands, as well as ensuring safety, and interacts well with the user.

Keywords—Guide dog robot, autonomous navigation, user command recognition, voice interaction

I. Introduction

According to data from the World Health Organization, approximately 292 million people worldwide have visual impairments, of which around 43 million are completely blind [1]. The mobility of visually impaired individuals primarily relies on guide dogs and white canes. However, guide dogs face challenges such as limited availability, high training costs, and a finite working lifespan [2]. White canes can only detect nearby obstacles and do not provide a comprehensive understanding of the surrounding environment. Moreover, individual needs vary in terms of travel speed, route selection, and navigation preferences, highlighting the necessity for a guidance system that can adapt to each user's specific requirements. Current guide dog robots can move together with users to the destinations, but it is still hard for the robot to understand the intention of the user [3-5]. Moreover, considerate functions are highly needed to improve the social acceptance of robots [6, 7].

To address these challenges, this paper develops a guide dog robot capable of operating in dynamic environments and responding to user commands. The system integrates autonomous navigation with user control via two physical buttons, allowing intuitive speed adjustments and pause/resume functionality. It also features a voice-based interface that delivers environmental descriptions, enhancing spatial awareness and user confidence. By combining these functions, the robot supports safe and flexible mobility according to individual needs. The proposed robot navigation system flowchart shows in Fig. 1.

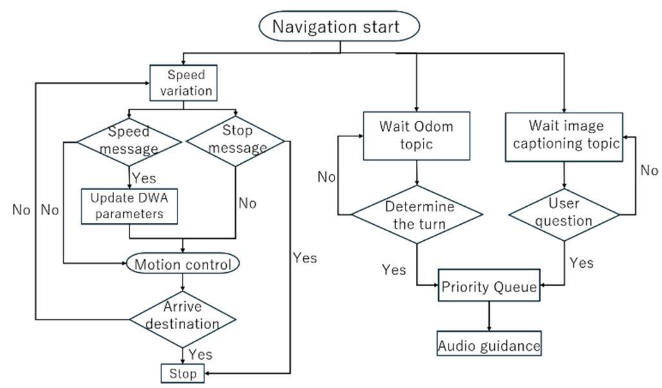


Fig. 1. Robot navigation system flowchart.

II. Autonomous Navigation

Autonomous mobility refers to the capability of robots to independently plan paths, avoid obstacles, and navigate within unknown or partially known environments. This is achieved through environmental perception using various sensors, which enable the construction of both global and local cost maps. Within the ROS move_base framework, the global cost map is utilized in conjunction with a global planner to generate an optimal path toward the target destination. Meanwhile, the local cost map, together with a local planner, is responsible for short-term path refinement and avoiding obstacles, producing motion commands that drive the robot's chassis to execute precise movements [8].

A. DWA Local planner

The Dynamic Window Approach (DWA) is a velocity-based local path planning method widely adopted in mobile robot navigation. Its core concept involves searching for the optimal velocity combination within the robot's feasible velocity space to ensure avoiding obstacles, smooth motion, and effective convergence toward the global path in complex environments. At the heart of DWA lies the dynamic window mechanism, which generates multiple candidate velocity trajectories based on the robot's current motion state and predefined velocity constraints. These candidates are then evaluated for feasibility within the velocity space.

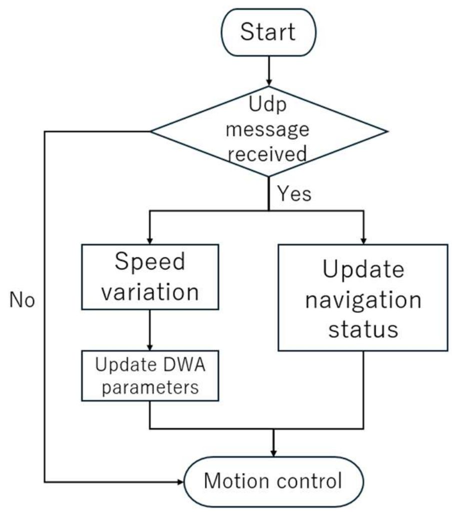


Fig. 2. Multi-robot system flowchart.

By incorporating the robot's kinematic model and real-time environmental perception data, DWA filters out trajectories that would result in collisions and selects the optimal one using a scoring function. This function typically comprises three components: path alignment, velocity cost, and obstacle distance. Path alignment encourages the robot to follow the global trajectory, the velocity term favors faster movement, and the obstacle distance ensures safety by penalizing proximity to obstacles.

DWA offers several advantages, including strong real-time performance, smooth trajectory generation, and adaptability to dynamic environments, making it a highly effective approach for local path planning in autonomous navigation.

However, DWA relies on predefined parameter settings, including upper limits for velocity. Consequently, the robot cannot exceed these bounds even in environments that permit higher speeds. This lack of dynamic acceleration and deceleration limits the robot's ability to adjust its speed based on real-time environmental changes or user command [9].

B. Configuration of the Handheld Device

To enable user interaction, two buttons are equipped to control robot speed and navigation state. A Raspberry Pi, programmed in Python, reads button signals via GPIO pins and transmits them to a host computer using UDP. A photocoupler ensures electrical isolation and stability in signal transmission.

C. Motion Control

As shown in Fig.2, the speed adjustment button is a momentary push button that returns to its original position after being pressed up or down. The robot's initial movement speed is set to 0.4 m/s, the system continuously listens for external commands; upon receiving a "+0.1" or "-0.1" input, it updates the max_vel_x parameter accordingly, ensuring that its value remains within the predefined range between min_vel_x and max_vel_x_limit to maintain safe operation. The updated max_vel_x is applied in real time using the rosrun dynamic_reconfigure dynparam set command, enabling move_base to reflect the new maximum speed immediately during navigation, allowing for three levels of acceleration and two levels of deceleration.

A second latching button starts or stops navigation using the ROS Action interface. The robot pauses when receiving command "0" and restarts with "1," sent over the same UDP channel. The move_base node acts as the Action server, processing goals and enabling flexible task execution.

This structure allows real-time, user-driven speed adjustment and navigation control while ensuring safe operation in dynamic environments.

III. VOICE GUIDANCE SYSTEM

The voice system operates in standby mode, awaiting predefined keywords. Upon detection, it provides voice descriptions of the environment in front of the robot and announces turning actions to enhance user awareness.

A. Image Captioning

Using the NIC (Neural Image Captioning) method, this system generates appropriate textual descriptions from images by combining deep learning and the attention mechanism, based on color images obtained from an RGB-D camera.

First, the system receives a color image from the camera and extracts its features using a CNN (Convolutional Neural Network) encoder. Next, these extracted features are fed into an LSTM (Long Short-Term Memory) decoder, which applies the attention mechanism to focus on different regions of the image while generating text step by step. Furthermore, Beam Search is utilized to select the most optimal description from multiple candidate sentences [10]. Additionally, OpenPose posture recognition is incorporated to determine whether a person in the image is standing or walking, and the generated description is adjusted accordingly based on the detection results [11, 12].

The generated image descriptions are then published to the caption_topic topic.

B. Voice Expressing

Built on the ROS operating system, the proposed system integrates speech recognition, text-to-speech synthesis, and robot navigation monitoring to enable voice announcements and interactive question-and-answer (Q&A) functionality. The system is shown in Fig. 3.

The system subscribes to the caption_topic to receive descriptive text and to the odom topic to monitor the robot's angular velocity. It utilizes the speech_recognition library for voice recognition, pyopenjtalk for text-to-speech conversion, and playsound for audio playback.

To manage voice output, the system employs a PriorityQueue to organize speech tasks, which are processed sequentially by a dedicated audio_player thread. This structure prevents redundant announcements and ensures smooth playback. To further reduce repetition, the remove_consecutive_duplicates method detects and removes any identical speech tasks that occur more than three times consecutively, ensuring that high-priority robot status messages are retained while minimizing excessive repetition.

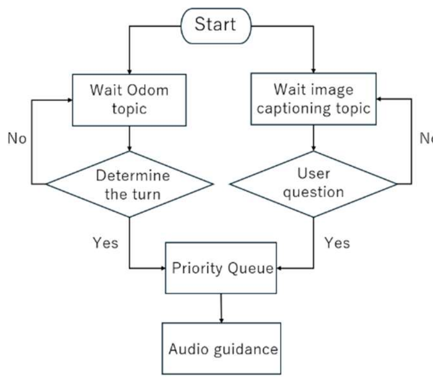
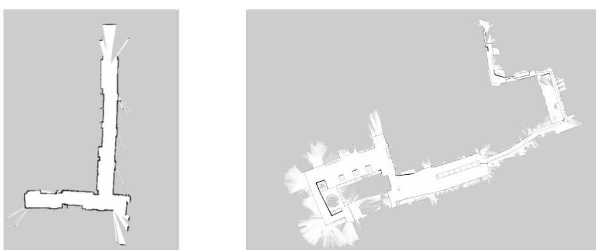


Fig. 3. Voice Guidance System flowchart.



(a)Indoor map (b)Outdoor map

Fig. 4. Environment map.

For turn detection, a monitor_turning thread continuously monitors the robot's angular velocity. A value of ≥ 0.48 rad/s is interpreted as a left turn, while ≤ -0.45 rad/s is recognized as a right turn. If the angular velocity returns to the neutral range of -0.05 to 0.05 rad/s, the turning status is cleared, enabling accurate detection of the next turning motion.

In Q&A mode, the system listens for user input via the speech_recognition library and converts spoken queries into text. Upon detecting relevant keywords, the system responds by playing the corresponding descriptive content from caption_topic, providing users with contextual environmental information. If no valid keyword is identified or speech recognition fails, the system prompts the user to repeat the query. It continues listening, thereby ensuring accurate interpretation of user command and delivering appropriate responses.

IV. Experiments

The indoor and outdoor experimental maps are shown in Fig. 4(a) and Fig. 4(b), respectively. In this study, multiple experiments were conducted in both indoor and outdoor environments to evaluate the effectiveness of the proposed system.

The indoor experiments included two comparative tests: one in a safe, obstacle-free environment, and another in an environment with obstacles. The obstacle-free test was designed to evaluate the robot's movement speed and operational behavior under ideal conditions. In contrast, the obstacle-present test aimed to assess whether the robot could appropriately adjust its speed by considering both the user's intent and environmental constraints.

(a)Safe environment (b)Environment with obstacles

Fig. 5. Indoor Environment

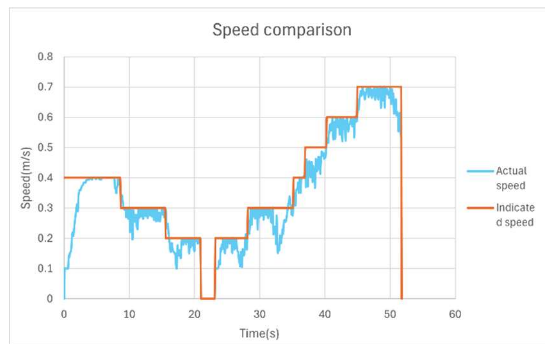


Fig. 6. Result under safe environment

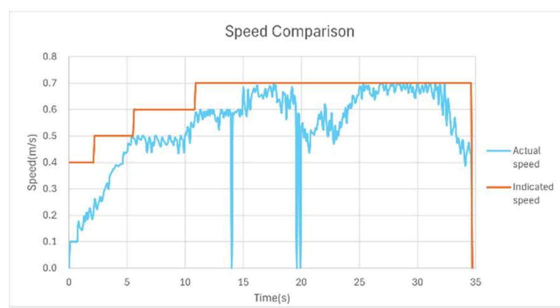


Fig. 7. Result under Obstacle environment

The outdoor experiment was conducted in a dynamic environment containing moving obstacles, with the objective of validating both the system's adaptability to real-world conditions and the performance of the voice guidance functionality.

A. Indoor Experiment

The safe indoor environment used in the experiment is shown in Fig. 5(a), and the corresponding results are presented in Fig. 6. In the graph, the red line represents the user-commanded movement speed, while the blue line indicates the robot's actual speed. The results indicate that, under obstacle-free conditions, the robot generally follows the user-specified speed accurately.

In contrast, the movement control experiment conducted in an environment with obstacles, as shown in Fig. 5(b), yielded the results presented in Fig. 7. In this scenario, the robot was

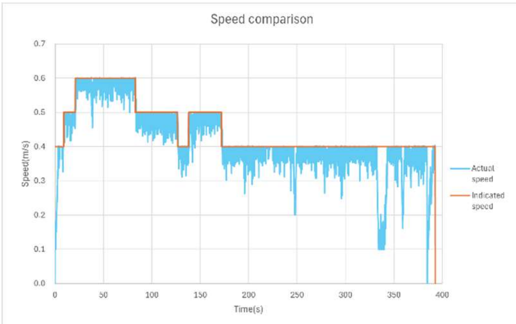


Fig. 8. Result under Obstacle environment

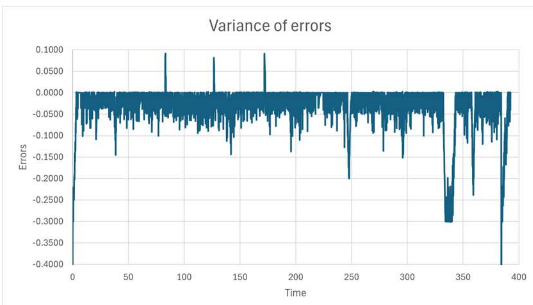
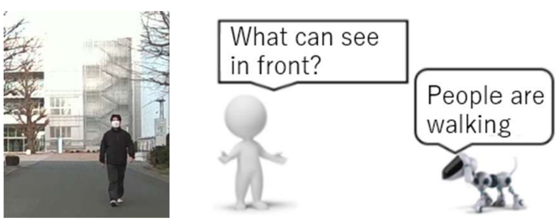


Fig. 9. Variance of errors



(a)Scene (b)Image captioning

Fig. 10. Result of voice guidance

able to accelerate smoothly in response to user input when positioned at a safe distance from obstacles. However, as it approached an obstacle, a noticeable reduction in speed was observed. This behavior illustrates that the robot actively reduces speed to ensure safety during avoiding obstacles. Once the obstacle was successfully bypassed, the robot resumed the user-defined speed. The robot modifies its speed based on environmental and user factors.

Overall, the experimental results validate that the proposed system enables safe and adaptive navigation. In particular, the speed modulation observed during avoiding obstacles indicates that the robot balances user input with safety in dynamic environments.

B. *Outdoor Environment*

The results of the outdoor experiment are shown in Fig. 8. The robot maintained a speed close to the user command while moving in straight paths but slowed down near obstacles due to dynamic changes in the environment. After passing the obstacles, the robot's speed returned to the commanded value.

Figure 9 presents the speed error analysis. The mean error was -0.0323 m/s, indicating that the actual speed was slightly lower than the commanded value. This deviation is mainly attributed to the following factors:

- The robot gradually accelerated from a stationary state, causing larger deviations during startup and stopping phases.
- The user adjusted speed in steps of ±0.1 m/s, but due to motor inertia and control latency, smooth transitions occurred instead of immediate changes.
- Environmental conditions such as uneven terrain and wind resistance also caused minor fluctuations.

To further improve system performance, alternative input methods may be explored, such as pressure sensors or guiding ropes, allowing for more intuitive and continuous speed control. Additionally, structural redesign and PID parameter optimization could enhance speed stability and control precision.

Fig. 10 presents the evaluation results of the voice Q&A system. During testing, the user wore a single-sided headset equipped with a noise-canceling function. Normally, the noise-canceling mode remained active to suppress background noise. When the user wished to ask a question, the mode was temporarily disabled, allowing speech input, and re-enabled afterward. This process was designed to minimize the impact of environmental noise and reduce unintended activation of the Q&A system. Upon detecting a keyword, the system responded with contextual information about the scene ahead, such as identifying whether a person was walking. Additionally, the robot provided turn-related notifications during directional changes.

Several issues were identified during Q&A operation. First, as the system relied on online processing, network latency affected the responsiveness of keyword detection. Second, manually toggling the mute function on the headset was somewhat inconvenient. To address these limitations, future improvements may include incorporating offline speech processing to reduce latency, implementing voice-activated wake-up mechanisms to eliminate manual operations, and replacing mechanical voice outputs with more natural-sounding synthesized speech to improve user experience.

V. CONCLUSION

This paper developed a guide dog robot that enables real-time speed adjustment based on user input and obstacle proximity, offering clear advantages in terms of flexibility and safety. Additionally, the robot is equipped with a voice interaction module. Environmental descriptions are automatically generated in real time by using LSTM neural networks, significantly enhancing users' spatial awareness and interaction experience in complex and dynamic environments.

Future work will focus on addressing the limitations of the current system and further improving its practicality, which includes enhancing adaptability to various environments, improving operational robustness and response speed, and exploring fully voice-based navigation control.

REFERENCES

[1] World Health Organization, *World Report on Vision*, Geneva, Switzerland: WHO, 2019. [Online]. Available: https://apps.who.int/iris/handle/10665/328717.

[2] B. Hong, Z. Lin, X. Chen, J. Hou, S. Lv, and Z. Gao, "Development and application of key technologies for guide dog robot: A systematic literature review," *Robotics and Autonomous Systems*, vol. 154, p. 104104, 2022, doi: 10.1016/j.robot.2022.104104.

[3] H. Hwang, T. Xia, I. Keita, K. Suzuki, J. Biswas, S. I. Lee, and D. Kim, "System Configuration and Navigation of a Guide Dog Robot: Toward Animal Guide Dog-Level Guiding Work," arXiv preprint arXiv:2210.13368, Oct. 2022. [Online]. Available: https://doi.org/10.48550/arXiv.2210.13368

[4] A. Xiao, W. Tong, L. Yang, J. Zeng, Z. Li, and K. Sreenath, "Robotic Guide Dog: Leading a Human with Leash-Guided Hybrid Physical Interaction," arXiv preprint arXiv:2103.14300, 2021. [Online]. Available: https://doi.org/10.48550/arXiv.2103.14300

[5] J. T. Kim, M. Byrd, J. L. Crandell, B. N. Walker, G. Turk, and S. Ha, "Understanding Expectations for a Robotic Guide Dog for Visually Impaired People," arXiv preprint arXiv:2501.04594, Jan. 2025. [Online]. Available: https://doi.org/10.48550/arXiv.2501.04594

[6] H. Hwang, T. Xia, I. Keita, K. Suzuki, J. Biswas, S. I. Lee, and D. Kim, "System configuration and navigation of a guide dog robot: Toward animal guide dog-level guiding work," arXiv preprint arXiv:2210.13368, 2022. [Online]. Available: https://doi.org/10.48550/arXiv.2210.13368

[7] H. Hwang, K. Suzuki, N. A. Giudice, J. Biswas, S. I. Lee, and D. Kim, "Lessons learned from developing a human-centered guide dog robot for mobility assistance," arXiv preprint arXiv:2409.19778, Sep. 2024. [Online]. Available: https://doi.org/10.48550/arXiv.2409.19778

[8] M. Quigley et al., "ROS: an open-source Robot Operating System," in *ICRA Workshop on Open Source Software*, Kobe, Japan, 2009, vol. 3, no. 3.2, p. 5.

[9] J. H. Zhang, Q. Feng, A. D. Zhao, W. He, and X. Hao, "Local path planning of mobile robot based on self-adaptive dynamic window approach," *Journal of Physics: Conference Series*, vol. 1905, no. 1, p. 012019, Sep. 2020, doi: 10.1088/1742-6596/1905/1/012019.

[10] K. Xu, J. Ba, R. Kiros, K. Cho, A. Courville, R. Salakhutdinov, R. Zemel, and Y. Bengio, "Show, attend and tell: Neural image caption generation with visual attention," *arXiv preprint arXiv:1502.03044*, Feb. 2015.

[11] Z. Cao, T. Simon, S. E. Wei, and Y. Sheikh, "OpenPose: Realtime Multi-Person 2D Pose Estimation using Part Affinity Fields," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 43, no. 1, pp. 172-186, Jan. 2021, doi: 10.1109/TPAMI.2019.2929257.

[12] R. Ichikawa, B. Zhang, and K. Hayashi, "A study on guide dog robots capable of expressing visual environments through voice," *IEEJ Transactions on Electronics, Information and Systems*, vol. 143, no. 5, pp. 562-568, 2023.